

Bachelor in Computer Applications

Course Title: **Computer Graphics**

Code No: Semester: **5**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Describe z-buffer method for visible surface detection. State its limitations and how these limitations can be reduced.
2. What is the role of clipping? Plot the line with end points (20, 10) and (30, 18) using DDA.
3. List the rotation matrix in clockwise direction with respect to x-axis, y-axis and z-axis. Rotate the object (0, 0, 0), (2, 3, 0), (5, 4, 0) about the rotation axis $y = 4$.
4. Write procedure to render the polygon surface by using Phong shading model.
5. Explain about parametric cubic curve with its properties.
6. Digitize circle with center (0, 0) and radius = 6.
7. Explain the use of polygon tables for boundary representations.
8. Differentiate between raster scan and random scan display systems.
9. What is intensity attenuation? Describe the various types of projection.
10. Explain about OpenGL and call back functions.
11. How the geometric and attribute information of a 3D objects are stored for the object representations? Explain with suitable example.
12. Write short notes on: a) Ambient light b) Diffuse reflection